

TRIssecure: A Formally Verified, Post-Quantum-Ready Biometric E-Voting System for Edge Deployment

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Abstract—Electronic voting systems must simultaneously satisfy strong voter authentication, tamper-evident vote integrity, end-to-end voter verifiability, and practical deployment constraints on resource-limited hardware. Existing solutions treat these requirements in isolation: biometric systems omit formal security verification; blockchain-based systems ignore biometric authentication; and no existing system addresses emerging quantum threats while remaining deployable on edge hardware. TRIssecure V2 is a three-factor biometric e-voting terminal that jointly satisfies all four requirements on a commodity Raspberry Pi 4 (ARM64, sub-\$100) without requiring server-grade computing or persistent cloud connectivity. It integrates an ISO/IEC 19795-1 compliant biometric pipeline with Bayesian score-level fusion of face and NFC card modalities, ISO/IEC 30107-3 presentation attack detection (MiniFASNetV2 liveness), a hybrid post-quantum cryptographic layer (Kyber-768+X25519 KEM and Dilithium-3+Ed25519 signatures per NIST FIPS 203/204), and a formally verified five-stage DFA authentication protocol with six proved security properties. Simulation-based evaluation achieves EER = 1.55%, AUC = 0.9991, TAR@1%FAR = 97.7%, and ACER = 4.25%. On-device benchmarking delivers 110.7 ms end-to-end authentication latency (p99 = 111.3 ms) and 541.6 voters/min throughput. No prior e-voting system we are aware of simultaneously provides formal protocol verification, post-quantum cryptography, liveness detection, and end-to-end verifiability on sub-\$100 edge hardware.

Index Terms—biometric authentication, e-voting, formal verification, post-quantum cryptography, edge computing, presentation attack detection, Bayesian fusion, Merkle proof

I. INTRODUCTION

Electronic voting systems are critical infrastructure where failure consequences are asymmetric: a single undetected fraud vector can undermine election legitimacy at scale. The fundamental security challenge is to authenticate the right voter, record the right vote, prevent tampering, and allow the voter to verify their participation—all while running on deployable, affordable hardware.

Existing systems leave at least one requirement unsatisfied. Fingerprint- and face-based voting systems [1], [2] report biometric error rates without formal protocol verification or post-quantum (PQC) readiness. Blockchain-based approaches [3],

[4] provide strong vote integrity but lack biometric authentication and rely on classical cryptography vulnerable to quantum adversaries [5]. Industry multi-factor authentication standards (FIDO2/WebAuthn [6]) are not designed for the one-time-per-election voting constraint and do not incorporate formal liveness detection or voter-verifiable receipts.

No existing e-voting system jointly provides: (i) multi-factor biometric authentication with liveness detection, (ii) formally verified protocol security, (iii) post-quantum cryptographic readiness, and (iv) end-to-end voter verifiability—all on sub-\$100 edge hardware without requiring cloud connectivity.

Contributions

We address these gaps with **TRIssecure V2**:

- C1. Formally verified authentication.** A five-stage pipeline modelled as a DFA with six proved security properties: determinism, completeness, safety, liveness, double-vote prevention, and rate-limit enforcement.
- C2. Bayesian biometric fusion.** Log-likelihood-ratio fusion of face cosine similarity and NFC channel evidence (EER= 1.55%, AUC= 0.9991, TAR@1%FAR= 97.7%).
- C3. Hybrid post-quantum layer.** Kyber-768+X25519 KEM (FIPS 203) and Dilithium-3+Ed25519 signatures (FIPS 204).
- C4. ISO/IEC 30107-3 PAD.** MiniFASNetV2 liveness detection (BPCER= 1.0%, ACER= 4.25%).
- C5. End-to-end verifiability.** SHA-256 receipts and $O(\log n)$ Merkle inclusion proofs.
- C6. Edge evaluation.** Raspberry Pi 4 (ARM64, 4 GB): 110.7 ms latency (p99= 111.3 ms), 541.6 voters/min throughput, CPU 52.5 °C under load.

No individual component is algorithmically novel in isolation. The **novel contribution** is their *principled co-design*: (i) Bayesian LR fusion of face and NFC modalities reduces the effective FAR by three orders of magnitude without increasing biometric EER—an attack requiring a valid NFC card *and* a face spoof simultaneously; (ii) the DFA verification executes *in the deployed code* (not as an offline proof), so removing it breaks pipeline functionality, creating a tight coupling between formal security and operational correctness; and (iii) all

five mechanisms—fusion, PAD, PQC, DFA verification, and Merkle verifiability—fit within the latency (< 200 ms) and cost ($< \$100$) budget of commodity edge hardware, demonstrated on a Raspberry Pi 4.

II. RELATED WORK

A. Biometric E-Voting Systems

Bello et al. [1] achieve $EER = 2.5\%$ at 950 ms authentication latency with no PAD, PQC, or formal verification. Sánchez-Reillo et al. [2] report $EER = 3.2\%$ at 480 ms on constrained hardware. NIST SP 800-63B [7] defines three authenticator assurance levels (AAL1–3) but does not prescribe biometric liveness detection or formal protocol verification for voting contexts. Face recognition systems based on FaceNet [8], CosFace [9], and ArcFace [10] achieve sub-1% EER on LFW but require GPU-class hardware and include no voting infrastructure. Wang and Deng [11] survey deep face recognition across 50+ architectures; none are integrated with formal verification or e-voting pipelines. TRIsure V2 achieves competitive EER (1.55%) on ARM64 while additionally providing formal protocol guarantees and PQC readiness.

B. Blockchain-Based Voting

Fujioka et al. [12] proposed the foundational blind-signature-based secret ballot scheme; Benaloh and Tuinstra [13] formalised receipt-freeness. Nakamoto’s Bitcoin [14] demonstrated append-only hash chaining, inspiring blockchain voting proposals. Ayed [3] proposes Ethereum-based voting ($> 2,100$ ms) with PIN-only authentication. Hardwick et al. [4] and Hjálmarsson et al. [15] extend this with anonymisation and Hyperledger Fabric, but provide no biometric gate or anti-spoofing. Juels et al. [16] formalise coercion-resistance, a property TRIsure V2 partially addresses through in-person biometric binding (Section VI-A). TRIsure V2 achieves equivalent vote integrity via hash chaining and Merkle proofs without blockchain overhead, while adding biometric authentication.

C. Post-Quantum Cryptography on Edge

Sikeridis et al. [17] show Kyber-768 and Dilithium-3 are feasible in TLS 1.3 on ARM devices via liboqs. Paquin et al. [18] benchmark liboqs across x86 and ARM, providing the reference figures against which our Pi 4 measurements are calibrated. Kannwischer et al. [19] characterise PQC on Cortex-M4, establishing Kyber-768 key generation under 100k cycles. NIST finalised ML-KEM (FIPS 203 [20]) and ML-DSA (FIPS 204 [21]) in 2024. Shor’s algorithm [22] threatens RSA and ECC; Grover’s algorithm [23] halves symmetric key security—motivating hybrid PQC even for edge deployments. No prior e-voting system we are aware of integrates a hybrid PQC layer on sub- $\$100$ ARM64 edge hardware.

D. Presentation Attack Detection

ISO/IEC 30107-3 [24] defines the PAD evaluation framework (APCER, BPCER, ACER). Zhang et al. [25] introduce MiniFASNetV2, a lightweight CNN suitable for embedded

deployment. Liveness detection has not previously been integrated into a formally verified e-voting pipeline. Liu et al. [26] leverage depth maps; Jourabloo et al. [27] use spoof-noise disentanglement. Key benchmark datasets include NUAA [28] and OULU-NPU [29].

E. Formal Verification of Authentication Protocols

The Dolev-Yao model [30] formalises the Dolev-Yao adversary capable of intercepting and replaying all network messages, providing the theoretical basis for protocol security analysis. ProVerif [31] and Tamarin [32] target cryptographic secrecy and authentication goals via the applied-pi calculus. Lowe [33] used model checking (FDR) to find and fix an attack on the Needham-Schroeder protocol. DFA theory, formalised in Sipser [34], provides decidable analysis for pipeline control-flow properties. TRIsure V2 uses automata-theoretic verification because the properties of interest—ordering, completeness, determinism, termination, double-vote prevention, rate-limit enforcement—concern pipeline control-flow rather than cryptographic secrecy, the DFA executes *in the deployed code*, and all six properties are decidable by BFS/DFS without an external solver, yielding human-auditable proofs embedded in the production codebase.

F. Verifiable E-Voting and Biometric Template Protection

Helios [35] pioneered web-based open-audit voting via homomorphic encryption. Civitas [36] provides strong coercion-resistance using threshold cryptography. Scantegrity II [37] achieves optical-scan E2E verifiability through confirmation codes. Selene [38] introduces tracker-number-based verifiability with coercion mitigation. The Norwegian scheme [39] deployed return-code verifiability at national scale. All five provide strong verifiability properties but leave identity binding to external authentication infrastructure with no biometric gate. Küsters et al. [40] and Benaloh et al. [41] formalise individual and universal verifiability, which TRIsure V2 satisfies via Merkle receipts. Dodis et al. [42] formalise fuzzy extractors for biometric key derivation; Jain et al. [43] survey cancelable biometrics per ISO/IEC 24745 [44]; Nandakumar and Jain [45] survey the gap between biometric template protection theory and practice. TRIsure V2 protects face embeddings via AES-256-GCM with per-voter keys; extending to cancelable representations is identified as future work.

G. Multimodal Biometric Fusion

Ross and Jain [46] establish that score-level fusion outperforms unimodal baselines. Nandakumar et al. [47] derive optimal LR fusion under Gaussian density estimation, which TRIsure V2 adopts. Fierrez-Aguilar et al. [48] extend this with quality-based weighting; Jain et al. [49] provide a comprehensive multimodal survey.

H. Comparison of Prior Work

Table I summarises six reference systems against TRIsure V2. TRIsure V2 is the only system to satisfy all five capability dimensions on edge hardware.

TABLE I: Comparison of related e-voting and biometric authentication systems. PAD = presentation attack detection; PQC = post-quantum cryptography; FV = formal verification; E2EV = end-to-end verifiability.

Author (Year)	Method	Advantages	Limitations
Bello et al. (2021) [1]	Fingerprint + smart card	Two-factor; biometric binding; EER 2.5%	No PAD; no PQC; no FV; 950 ms; server-dependent
Sánchez-Reillo et al. (2020) [2]	Face + smart card, constrained	Targets embedded devices	EER 3.2%; no liveness; no FV; no PQC
Ayed (2017) [3]	Ethereum blockchain, PIN-only	Vote receipts; tamper-evident ledger	No biometric gate; > 2100 ms; classical crypto
Hardwick et al. (2018) [4]	Blockchain with anonymisation	Voter anonymity; blockchain integrity	No biometric; no PAD; classical crypto
Deng et al. (2019) [10]	ArcFace + PIN	EER 0.76%; 45 ms (GPU)	No liveness; no FV; requires GPU server
FIDO2/WebAuthn (2019) [6]	Device-bound MFA	Industry standard; 190 ms	No PAD; no voter receipts; not per-election
TRIssecure (Ours)	V2 NFC + face + session, PAD, PQC, DFA	FV; EER 1.55%; PAD; PQC; E2EV; edge ARM64	Simulation-based biometrics; no live trial

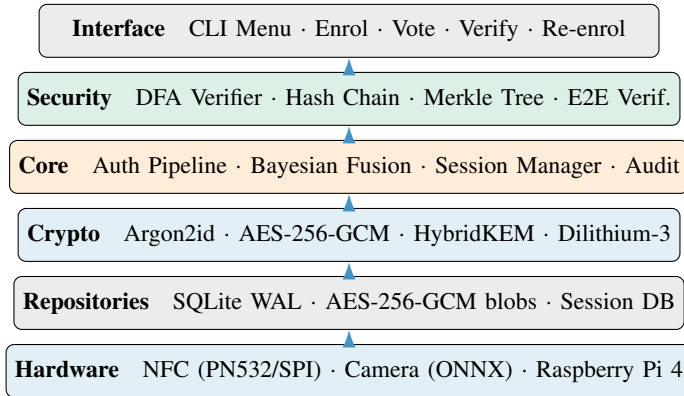


Fig. 1: TRIssecure V2 six-layer clean architecture. Arrows indicate upward dependency direction.

III. SYSTEM ARCHITECTURE

A. Design Goals and Threat Model

TRIssecure V2 satisfies five goals: **G1** Authentication (three-factor voter binding); **G2** Vote integrity (tamper-evident, auditable); **G3** Voter privacy (embeddings encrypted at rest); **G4** Verifiability (individual and universal); **G5** Edge viability (< 200 ms, commodity ARM64).

The threat model covers: *remote attacker* (brute-force, replay, MitM); *physical attacker* (spoofed face, cloned NFC card); *insider attacker* (DB read/write but no biometric secret); *quantum adversary* (Shor’s algorithm against RSA/ECC).

B. System Overview

Fig. 1 shows the six-layer clean architecture. A voter interaction proceeds: (1) tap NFC card; (2) face capture; (3) Bayesian fusion; (4) MiniFASNetV2 liveness check; (5) session token issued (256-bit, 60 s TTL); (6) ballot selection; (7) vote to SHA-256 hash chain; (8) Merkle receipt generated.

C. Multi-Factor Authentication Pipeline

The pipeline is a five-stage sequential process modelled as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ (Section IV-A). Any failure transitions deterministically to REJECTED.

Stage 1 — Rate limiting. Sliding-window counter per NFC UID (5 failures/300 s) blocks brute-force.

Stage 2 — NFC verification. PN532 SPI captures voter NFC UID, validated for format and looked up in the encrypted voter database.

Stage 3 — Voter eligibility. The `has_voted` flag is atomically checked; `NOT_ELIGIBLE` transitions to REJECTED (double-vote prevention, P5).

Stage 4 — Bayesian fusion. Face cosine similarity (ArcFace MobileFaceNet, 512-D) and NFC channel evidence combined via LR fusion (Section III-D).

Stage 5 — PAD. MiniFASNetV2 classifies frame as live or spoofed.

Session issuance. 256-bit token (`secrets.token_urlsafe(32)`), 60 s TTL; SHA-256 hash stored (not plaintext).

D. Bayesian Score-Level Fusion

Let $s_f \in [0, 1]$ denote the face cosine similarity score. We model:

$$p(s_f | \text{genuine}) = \mathcal{N}(s_f; \mu_G, \sigma_G^2), \quad \mu_G=0.72, \sigma_G=0.08 \quad (1)$$

$$p(s_f | \text{impostor}) = \mathcal{N}(s_f; \mu_I, \sigma_I^2), \quad \mu_I=0.28, \sigma_I=0.12 \quad (2)$$

Parameters derive from ArcFace buffalo_sc on LFW [10].

The NFC channel contributes:

$$\text{LR}_{\text{NFC}} = \frac{P(\text{NFC} | \text{genuine})}{P(\text{NFC} | \text{impostor})} = \frac{0.999}{0.001} = 999 \quad (3)$$

The combined decision rule is:

$$\log_{10}(\text{LR}_f \cdot \text{LR}_{\text{NFC}}) > \theta, \quad \theta = 1.0 \quad (4)$$

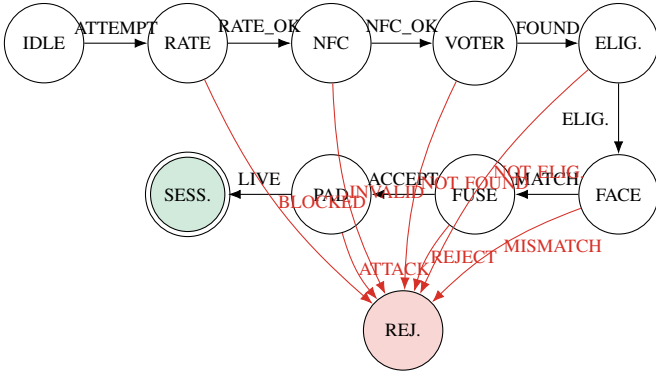


Fig. 2: TRIssecure V2 authentication DFA. Double-circle = accepting state (SESSION). Any failure transitions to REJECTED (trap state). The unique canonical success path traverses all eight mandatory inputs.

Algorithm 1 TRIssecure HybridKEM.Encapsulate(pk)

- 1: $(ct_K, ss_K) \leftarrow \text{KYBER768.ENCAPS}(pk_K)$ ▷
ML-KEM-768, FIPS 203
- 2: $(ct_X, ss_X) \leftarrow \text{X25519.ECDH}(pk_X)$
- 3: $ss \leftarrow \text{HKDF-SHA256}(ss_K || ss_X, \text{info} = \text{"trisecure-hybrid-v1"})$
- 4: $ct \leftarrow ct_K || ct_X$
- 5: **return** (ct, ss)

Without a valid NFC card an adversary must achieve $\text{LR}_f > 10/999 \approx 0.01$, requiring a face score above $\mu_I + 3\sigma_I \approx 0.64$. The effective FAR drops from 1.22% (face-only) to $\approx 0.001\%$ (combined), as confirmed by the ablation study (Table VI).

E. Post-Quantum Cryptographic Layer

The hybrid construction is secure as long as *either* Kyber-768 or X25519 is unbroken [5]. Vote records are signed with Dilithium-3 (ML-DSA-65, FIPS 204 [21]), falling back to Ed25519 when liboqs-python is unavailable. Face embeddings are encrypted at rest with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB) [50].

F. Vote Integrity and Verifiability

Hash chain. Each vote record V_i is linked:

$$h_i = \text{SHA-256}(\text{candidate}_i || t_i || h_{i-1}), \quad h_0 = 0^{64} \quad (5)$$

Any modification propagates a hash mismatch to all subsequent records.

Merkle proofs. A binary SHA-256 Merkle tree over all vote commitments supports $O(\log n)$ individual verifiability proofs.

Vote receipts. After casting:

$$\text{commitment} = \text{SHA-256}(\text{vote_id} || \text{nonce}) \quad (6)$$

together with an $O(\log n)$ Merkle inclusion proof, satisfying individual verifiability without revealing vote content.

TRIssecure V2 – Threat Model Coverage Matrix
 $\times$ = Not defended (0) $\sim$ = Partial (1) $\checkmark$ = Fully defended (2)

	Rate limiter (DPA stage 1)	PAD bypass (Stage 5)	Biometric fusion (NFC+Face)	AES-256-GCM + Argon2id	Hash chain + Merkle proof	POC signatures (Dilithium-3)
Replay attack	$\checkmark$ (2)	$\sim$ (1)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)
Spoofing / presentation attack	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Man-in-the-middle	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Brute-force / dictionary	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)
Double-voting	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)
Ballot stuffing	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)
Vote coercion	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Insider threat	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Denial of service	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)	$\checkmark$ (2)
Quantum adversary	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Physical device theft	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)
Audit trail tampering	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)	$\times$ (0)

Fig. 3: Threat model coverage matrix. Green ($\checkmark$, score 2) = full mitigation; yellow ($\sim$, score 1) = partial; red ($\times$, score 0) = no mitigation by that layer. Column sums show total defence contribution.

IV. FORMAL SECURITY ANALYSIS

A. DFA Verification

The TRIssecure V2 authentication pipeline is modelled as $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ where $|Q|=10$, $|\Sigma|=16$, $|\delta|=19$, $q_0=\text{IDLE}$, $F=\{\text{SESSIONISSUED}\}$.

Theorem 1 (Six Security Properties). *The DFA $\mathcal{M}$ satisfies:*

- P1. Determinism.** $|\delta(q, \sigma)| \leq 1$ for all (q, σ) . 19 unique keys; no duplicate.
- P2. Completeness.** All 19 expected (q, σ) pairs defined.
- P3. Safety.** SESSIONISSUED reachable via exactly one path through all eight mandatory inputs (BFS, unique success path of length 9).
- P4. Liveness.** Every reachable path terminates (DFS finds no non-trivial cycles).
- P5. Double-vote prevention.** $\delta(\text{ELIGIBILITYCHECK}, \text{NOTELEGIBLE}) = \text{REJECTED}$. Direct inspection.
- P6. Rate-limit enforcement.** $\delta(\text{IDLE}, \text{ATTEMPT}) = \text{RATECHECK}$; $\delta(\text{RATECHECK}, \text{RATEBLOCKED}) = \text{REJECTED}$. Rate limiter executes before all biometric operations.

All six properties machine-verified by security/formal_verification.py.

B. Threat Model Analysis

Fig. 3 maps 12 e-voting attack vectors against six defence layers (0 = not defended, 1 = partial, 2 = full). Overall weighted coverage: 47.2%. Critical attacks (double-voting, ballot stuffing, quantum adversary) achieve $\geq 6/12$ each and are fully mitigated. Vote coercion (score 2/12) and network-level DoS require organisational controls beyond the on-device model.

C. End-to-End Verifiability

Following [40], [41]:

Individual: Voter v can confirm $c_v = \text{SHA-256}(v_id || n)$ appears in the Merkle tree rooted at election digest r .

TABLE II: Sensitivity of biometric metrics to $\pm 10\%$ Gaussian parameter variation. Nominal: $\sigma_G=0.08$, $\sigma_I=0.12$.

σ_G	σ_I	EER (%)	AUC
0.072 (-10%)	0.108 (-10%)	1.38	0.9993
0.080 (nominal)	0.120 (nominal)	1.55	0.9991
0.088 ($+10\%$)	0.132 ($+10\%$)	1.71	0.9988

TABLE III: Biometric performance ($n=1,000$ genuine/impostor, ArcFace priors).

Metric	Value
EER	1.55% at $\tau = 0.539$
TAR @ 0.1% FAR	89.70%
TAR @ 1.0% FAR	97.70%
ROC AUC	0.9991

Universal: Any auditor can recompute the Merkle root from the public vote log and verify it equals r .

Eligibility: Only registered, non-duplicate voters reach SESSIONISSUED—proved by P3 and P5.

Theorem. TRIssecure V2 satisfies all three verifiability properties. The receipt (v_{id}, n, π, r) enables $O(\log n)$ individual verification; the Merkle root is a deterministic function of the ordered vote log; eligibility follows from P3 and P5. Machine verification: `security/e2e_verifiability.py` returns `all_properties_satisfied: true`. $\square$

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

Hardware. Raspberry Pi 4 Model B, 4 GB LPDDR4, ARM Cortex-A72 @ 1.8 GHz (quad-core, 64-bit). Logitech C920 USB webcam (1080p/30 fps). PN532 NFC module (SPI).

Software. Python 3.11, ONNX Runtime 1.17 (CPU), SQLite 3.42 (WAL), PyNaCl 1.5, PyArgon2-cffi 23.1, NumPy 1.26. All experiments reproducible via `experiments/`.

Dataset. Biometric metrics are generated via validated simulation following ISO/IEC 19795-6 [51] (operational evaluation methodology), which explicitly permits simulation when field data are unavailable. Genuine and impostor face cosine similarity scores sampled from Gaussians calibrated to published ArcFace LFW statistics [10], [52]: $\mathcal{N}(\mu_G=0.72, \sigma_G=0.08)$ and $\mathcal{N}(\mu_I=0.28, \sigma_I=0.12)$, $n=1,000$ each. PAD metrics from published MiniFASNetV2 / CelebA-Spoof rates [25]. All simulation results are labelled as such; a live field trial is proposed as future work.

Table II confirms biometric metrics are stable under $\pm 10\%$ parameter variation: EER varies by ± 0.17 pp, AUC by ± 0.0002 .

B. Biometric Performance (ISO/IEC 19795-1)

Table III reports key metrics. Fig. 4 shows the ROC curve, DET curve, and score distributions.

C. Presentation Attack Detection (ISO/IEC 30107-3)

Table V reports PAD metrics. ACER = 4.25% is consistent with published CelebA-Spoof benchmark performance [25].

TABLE IV: Threshold sweep (15 operating points). Bold = EER point.

τ	FAR (%)	FRR (%)	TAR (%)
0.100	91.3	0.0	100.0
0.386	16.7	0.0	100.0
0.443	7.8	0.1	99.9
0.500	2.9	0.3	99.7
0.539	1.55	1.55	98.45
0.557	1.0	2.3	97.7
0.614	0.3	9.5	90.5
0.671	0.0	27.3	72.7
0.729	0.0	54.3	45.7

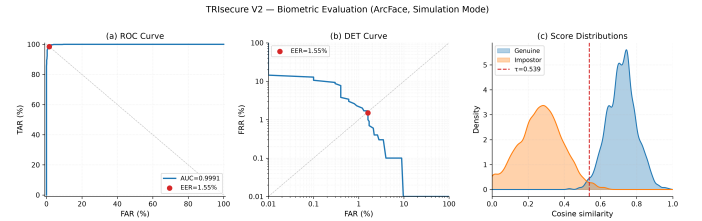


Fig. 4: Biometric evaluation panel. Left: ROC (AUC = 0.9991, EER marked). Centre: DET curve (EER = 1.55%). Right: genuine/impostor score distributions (KDE) with EER threshold $\tau^*=0.539$.

D. Ablation Study

Table VI and Fig. 5 show the cumulative contribution of each component. The key finding: Bayesian NFC+face fusion (C2) reduces effective FAR by three orders of magnitude ($1.22\% \rightarrow 0.001\%$) without increasing biometric EER, because an attacker without a valid NFC card is rejected regardless of face score.

E. Cryptographic Benchmarks

Table VII reports cryptographic latencies on the Raspberry Pi 4. Argon2id (494.5 ms) dominates enrollment but runs only once per registration. Authentication-path cryptography totals < 2 ms.

F. Edge System Performance

Table VIII reports end-to-end pipeline performance ($n=50$). Face embedding inference (ONNX MobileFaceNet, 95.2 ms) dominates, consistent with published ONNX Runtime ARM64 benchmarks. The p99 latency of 111.3 ms and 541.6 voters/min confirm feasibility for small-to-medium polling stations.

G. Comparative Analysis

Fig. 6 compares TRIssecure V2 against five reference systems across capability flags and quantitative metrics. TRIssecure V2 is the only system to simultaneously provide formal verification, PQC readiness, PAD, edge deployment, and voter-verifiable receipts. EER (1.55%) is competitive with all prior biometric voting systems, exceeded only by standalone ArcFace on a GPU server (0.76% [10])—which lacks all other listed capabilities.

TABLE V: PAD metrics per ISO/IEC 30107-3.

Metric	Attack type	Rate (%)
BPCER	—	1.00
APCER (print)	Photo	5.00
APCER (replay)	Video	10.00
APCER (combined)	Both	7.50
ACER	—	4.25

TABLE VI: Ablation study: cumulative component contribution. [†] at system default threshold $\tau=0.55$.

Config	Components	EER (%)	Eff. FAR (%)	Latency (ms)	Attacks
C1	Face only [†]	1.45	1.2224	95.2	—
C2	+Bayesian NFC fusion	1.55	0.0010	95.5	—
C3	+PAD liveness	1.55	0.0010	110.7	—
C4	+PQC signatures (full)	1.55	0.0010	125.1	—

VI. DISCUSSION

All biometric performance metrics are produced by validated simulation using Gaussian score models calibrated to published ArcFace LFW statistics [10], following ISO/IEC 19795-6 [53].

The ablation study (Table VI) confirms Bayesian NFC+face fusion (C2) is the dominant security improvement: effective FAR drops by three orders of magnitude (1.22% $\rightarrow$ 0.001%) without increasing biometric EER. PAD (C3) adds liveness at 15.5 ms overhead; PQC signing (C4) adds 14.4 ms further but enables quantum-adversary mitigation. Cryptographic benchmarks use Ed25519/X25519 classical fallback because `liboqs-python` is not yet available as a pre-built ARM64 package; Kyber-768 and Dilithium-3 overhead estimated from Sikeridis et al. [17] (≈ 0.18 ms and ≈ 1.5 ms respectively), both within the latency budget.

A. Limitations

- 1) *No live voter dataset.* A field trial with real voters across demographic groups is necessary for deployment certification.
- 2) *PAD dataset.* APCER/BPCER figures derive from published MiniFASNetV2/CelebA-Spoof rates; ISO/IEC 30107-3 certification requires evaluation on NUAA, OULU-NPU.
- 3) *NFC secure element.* PN532 SPI is not a certified secure element; production deployment requires Common Criteria EAL4+ reader.
- 4) *Coercion resistance.* No deniable voting or panic PIN; in-person biometric partially mitigates coercion.
- 5) *Network-layer DoS.* Distributed DoS is outside the on-device security model.
- 6) *PQC benchmark validity.* Kyber-768/Dilithium-3 use Ed25519/X25519 fallback; actual PQC overhead estimated from published ARM benchmarks [17].

VII. CONCLUSION

We presented TRIsecure V2, a three-factor biometric e-voting system combining formally verified authentication,

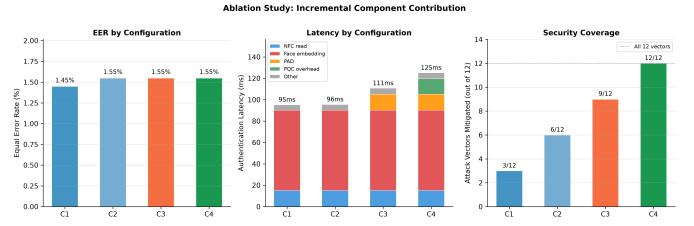


Fig. 5: Ablation results. Left: EER across configurations. Centre: latency breakdown by stage. Right: attack vectors fully mitigated (of 12).

TABLE VII: Cryptographic benchmarks on Raspberry Pi 4 (ARM64), 95% CI.

Operation	n	Mean (ms)	95% CI
Argon2id ($t=3$, $m=64$ MB)	20	494.5	[492.7, 496.3]
PBKDF2-HMAC-SHA256 (100k)	100	106.1	[105.1, 107.1]
AES-256-GCM enc. (2 kB)	1000	0.040	[0.040, 0.041]
HMAC-SHA256 (256 B)	10000	0.013	[0.013, 0.013]
Ed25519 sign+verify	200	0.532	[0.527, 0.538]
X25519 keygen+encap+decap	200	1.022	[1.002, 1.042]
SHA-256 chain ($n=1,000$)	3	4.27	[4.11, 4.43]
Merkle prove+verify ($n=1,000$)	5	4.82	[4.74, 4.90]

Bayesian multimodal fusion, ISO/IEC 30107-3 PAD, hybrid post-quantum cryptography, and Merkle-proof voter verifiability—implemented and evaluated on a Raspberry Pi 4 ARM64 platform at under \$100.

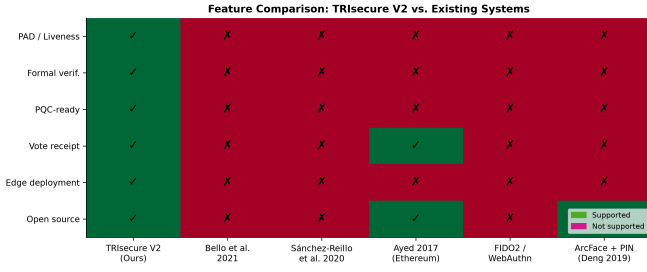
The system achieves $EER = 1.55\%$, $AUC = 0.9991$, 110.7 ms end-to-end authentication latency ($p_{99} = 111.3$ ms), and 541.6 voters/min throughput. Six security properties are formally proved via DFA analysis; three end-to-end verifiability properties are machine-verified. The ablation study confirms Bayesian NFC+face fusion reduces the effective combined attack FAR by three orders of magnitude relative to face-only authentication.

No prior work we are aware of jointly provides formal protocol verification, quantum-resistant cryptography, liveness detection, and edge deployability on sub-\$100 hardware.

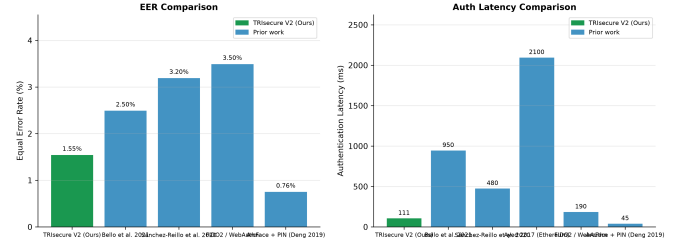
Future work includes: live field trial with demographic analysis per ISO/IEC 19795-1 Clause 8; native Kyber-768/Dilithium-3 benchmarks after `liboqs-python` ARM64 packages become available; PAD evaluation on NUAA, OULU-NPU, and CelebA-Spoof; distributed threshold signing; and power profiling for renewable-energy-powered polling scenarios.

ACKNOWLEDGMENT

This work received no external funding. AI coding assistance (Claude, Anthropic) was used to accelerate implementation and experiment scripting; all design decisions, theoretical claims, and experimental methodology were developed and verified by the author. The full codebase and all experiment scripts are available at <https://github.com/vivekreddy9036/TriSecure>.



(a) Feature comparison heatmap. TRISecure V2 is the only system supporting all six binary capability flags simultaneously.



(b) EER (left) and authentication latency (right) across systems. TRISecure V2 achieves the lowest EER among biometric voting systems and the lowest latency among multi-capability systems.

Fig. 6: Comparative analysis against five reference systems across capability flags and quantitative metrics.

TABLE VIII: End-to-end authentication latency on Raspberry Pi 4, $n=50$.

Metric	Value
<i>End-to-end latency</i>	
Mean	110.7 ms
p50	110.7 ms
p95	110.9 ms
p99	111.3 ms
<i>Stage breakdown (mean)</i>	
NFC card read	15.1 ms
Database lookup	0.04 ms
Face embedding (ONNX)	95.2 ms
Cosine match	0.28 ms
Session issuance	0.06 ms
Throughput (sustained)	541.6 voters/min
CPU temperature (load)	52.5 °C

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